Medipally Sanjit Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.71
Fiitjee Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	98.20%
Fiitjee School	Class X (Telangana State Board)	2021	10.00%

PROJECTS

Multithreading, Scheduling & Synchronization

C++ | *POSIX Threads* | *Concurrency* | *Synchronization* | *Atomic Operations*

- Developed a **parallel Sudoku validation system** using POSIX threads with Chunk, Mixed (cyclic), and Dynamic scheduling, evaluating workload distribution across varying thread counts and Sudoku sizes.
- Implemented **dynamic task allocation and mutual exclusion** using atomic **Test-and-Set (TAS), Compare-and-Swap (CAS), and Bounded CAS**, and analyzed synchronization overhead and scalability.
- Implemented the **bounded-buffer Producer-Consumer problem** using both **semaphores and locks**, and compared their performance across different producer/consumer delay and thread-count ratios.

Virtual Memory Management in gemOS

C | *Operating Systems* | *Virtual Memory* | *Page Tables*

- Implemented **virtual memory area (VMA) management** through mmap(), munmap(), and mprotect(), handling VMA creation, deletion, splitting, merging, and protection changes.
- Implemented **lazy page allocation and page-fault handling**, validating memory accesses against VMA permissions and allocating physical page frames on first access.
- Built and updated **multi-level page-table mappings** for newly allocated pages, configuring page-table entries according to the access permissions of the corresponding virtual memory region.
- Extended munmap() and mprotect() to update virtual-to-physical mappings and enforce modified memory-access permissions.

Mini-ML Library

Python | *NumPy* | *Machine Learning* | *Neural Networks*

- Built a **modular machine-learning library from scratch** with a scikit-learn-style API, implementing preprocessing utilities and **L2-regularized Logistic Regression using IRLS/Newton-Raphson**.
- Developed a **mini automatic differentiation engine** with computational-graph construction and backward propagation for arithmetic, matrix multiplication, activation and reduction operations.
- Implemented core neural-network components including **Linear, ReLU, Sigmoid, Sequential, CrossEntropyLoss and SGD**, and evaluated multiple models on **Fashion-MNIST**.

Chemical Process Multi-Objective Optimization

Python | *TensorFlow/Keras* | *ANN* | *NSGA-II* | *pymoo*

- Developed an end-to-end data-driven optimization framework for a simulated chemical reactor using **10,000 records, 6 process variables, and yield/impurity as competing objectives**.
- Built and trained a **64-32-16 MLP** with ReLU activations using TensorFlow/Keras, Adam optimization, MSE loss, and early stopping, achieving **R^2 0.91** on training, validation, and test sets.
- Integrated the trained ANN with **NSGA-II using pymoo** to perform multi-objective optimization and identify **Pareto-optimal process configurations** balancing reactor yield and impurity.
- Analyzed the resulting **Pareto front and trade-offs between objectives**, identifying process-variable combinations that provide different compromises between maximizing yield and minimizing impurity.

SKILLS

- **Programming:** C, C++, Python, SQL
- **Computer Science**: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems
- **Libraries**: NumPy, SciPy, Matplotlib, Pandas, TensorFlow/Keras, Scikit-learn
- **Tools**: Git, GitHub, Linux, VS Code, Jupyter Notebook

SCHOLASTIC ACHIEVEMENTS

- Achieved a state rank of 5 in NTSE stage 1 examination conducted in the year 2021.
- Qualified UGEE 2023, IIIT Hyderabad and got shortlisted for the interview round.

RELEVANT COURSES

- **Computer Science Courses:** Introduction to Programming(C/C++), Data Structures & Applications, Operating Systems I & II, Foundations of Machine Learning, Optimization Techniques-I, ML for BioInformatics
- **Math Courses**: Calculus, Linear Algebra, Probability, Transform Techniques, Differential Equations, Complex Variables, Statistics
- **Numerical Methods Courses**: Fundamentals of Scientific Computing, Computer Aided Numerical Methods, Computational Methods in Material Science
- **Soft Skills Courses**: Communication Skills, Ethics and Values, Intro To Entrepreneurship